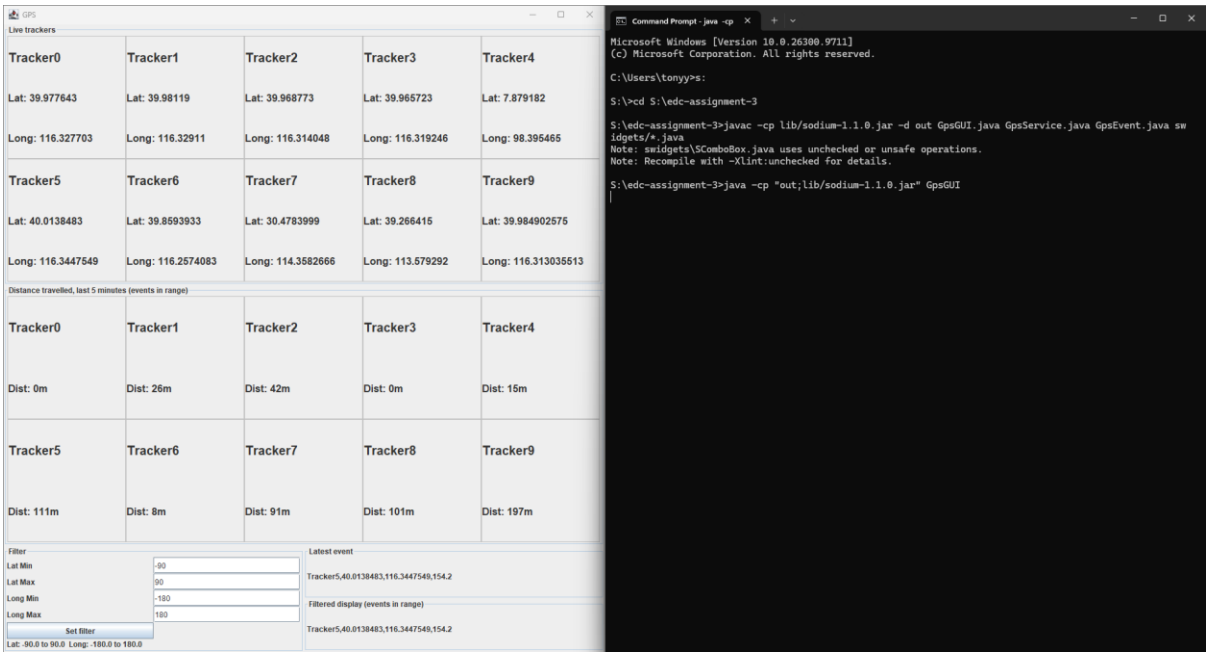
How to run the program

javac -cp lib/sodium-1.1.0.jar -d out GpsGUI.java GpsService.java GpsEvent.java swidgets/*.java

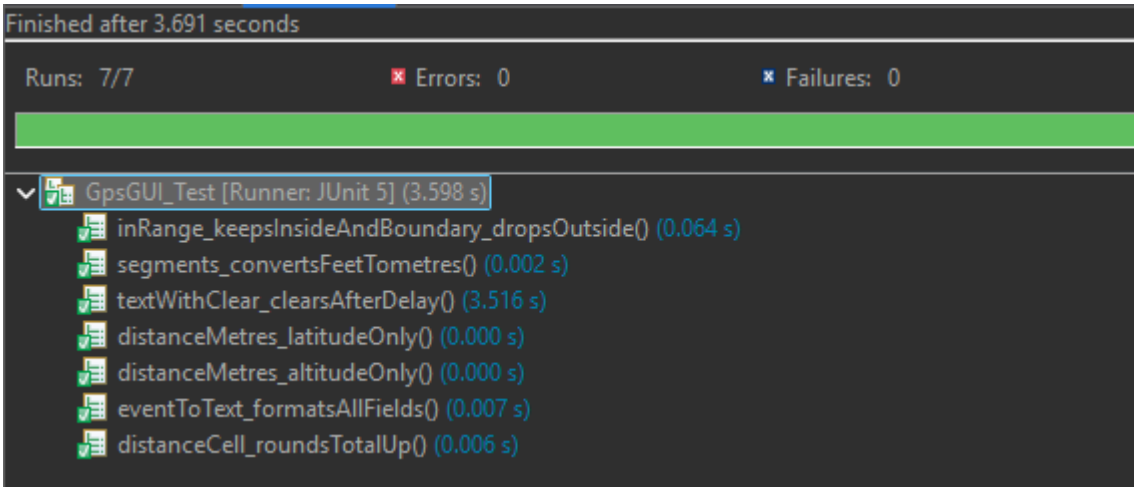
java -cp "out;lib/sodium-1.1.0.jar" GpsGUI



How to run the unit test

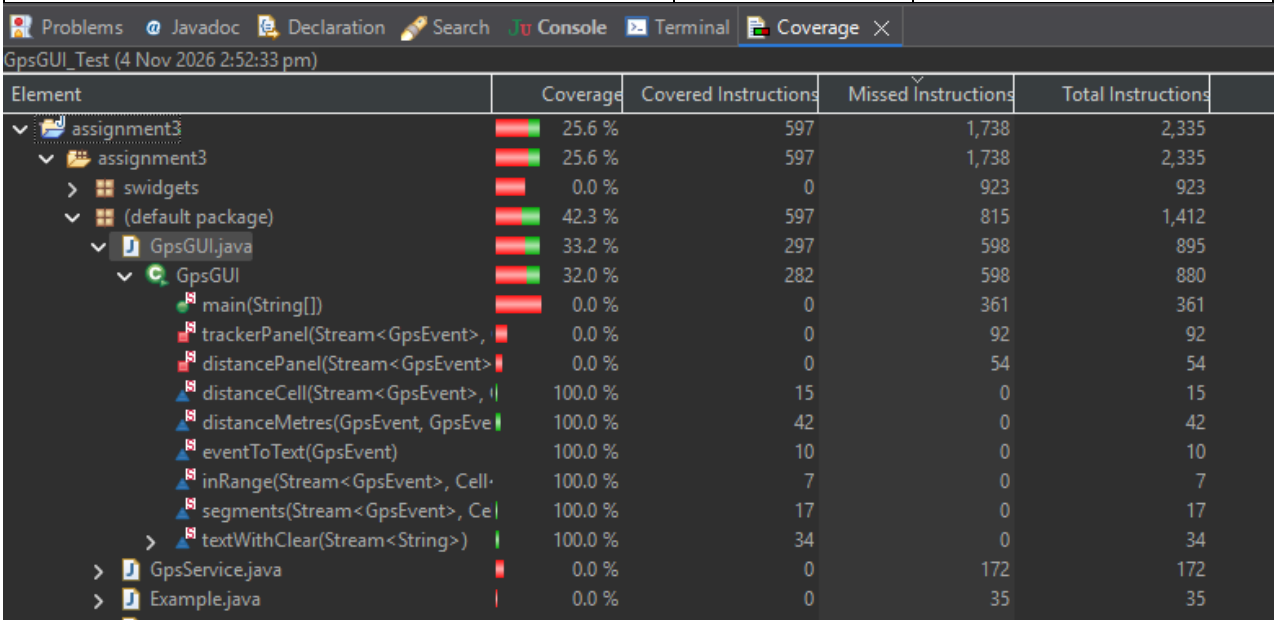
What I ran:

1. In Eclipse, right-click GpsGUI_Test.java, then Run As, then Junit Test.
2. For coverage: right-click it, then Coverage As, then Junit Test.

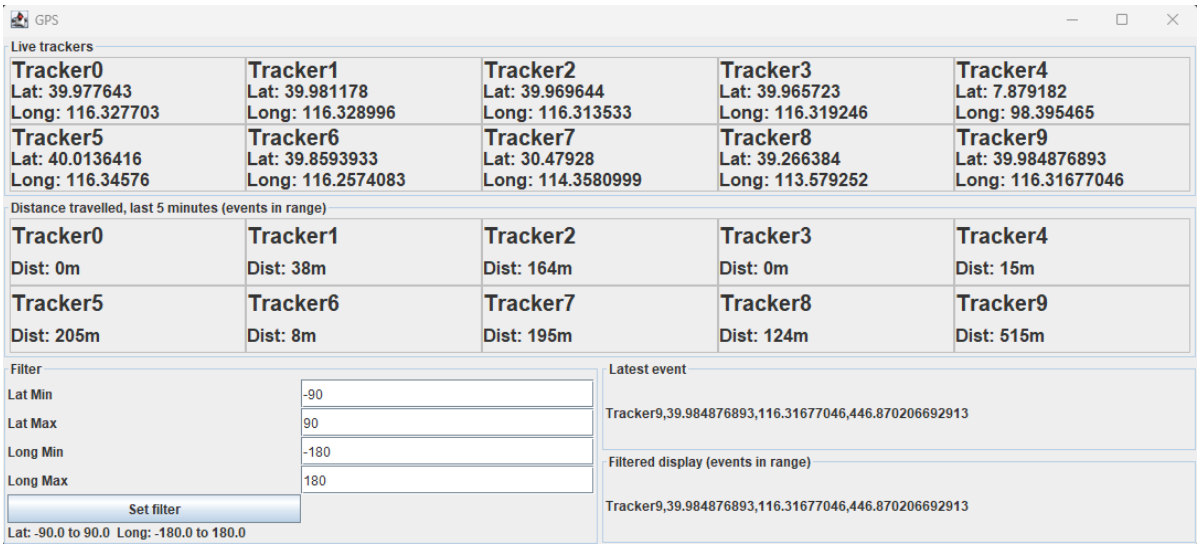


What each test checks:

Test	What it checks	Expected result
distanceMetres_latitudeOnly	Two points 0.001 of latitude apart	about 111.32m
distanceMetres_altitudeOnly	Two points 10m apart in altitude only	10
eventToText_formatsAllFields	Text format	Tracker1,10.5,20.5,30.5
inRange_keepsInsideAndBoundary_dropsOutside	Filter that keeps the events inside the range and on its boundary, and drops event outside it.	2 of the events pass
Segments_convertsFeetTometres	Converts altitude from feet to metres and the first event produce no segment	1 segment of about 30.48m
distanceCell_roundsTotalUp	Distance label shows the total in metres, rounded up to the next whole number	“Dist: 31m”
textWithClear_clearsAfterDelay	Text shows when it arrives and it’s cleared after about 3 seconds if nothing new comes in.	“waiting…” at the start, “hello” right after the event, “” after the wait.



GpsGUI is at 33.2%. The six helper methods that hold the logic (distanceMetres, eventToText, inRange, segments, distanceCell, textWithClear) are at 100%. Main, trackerPanel and distancePanel only build the window, so they are covered by the GUI test below.



Started the program from the terminal and waited until all ten trackers had received an event.

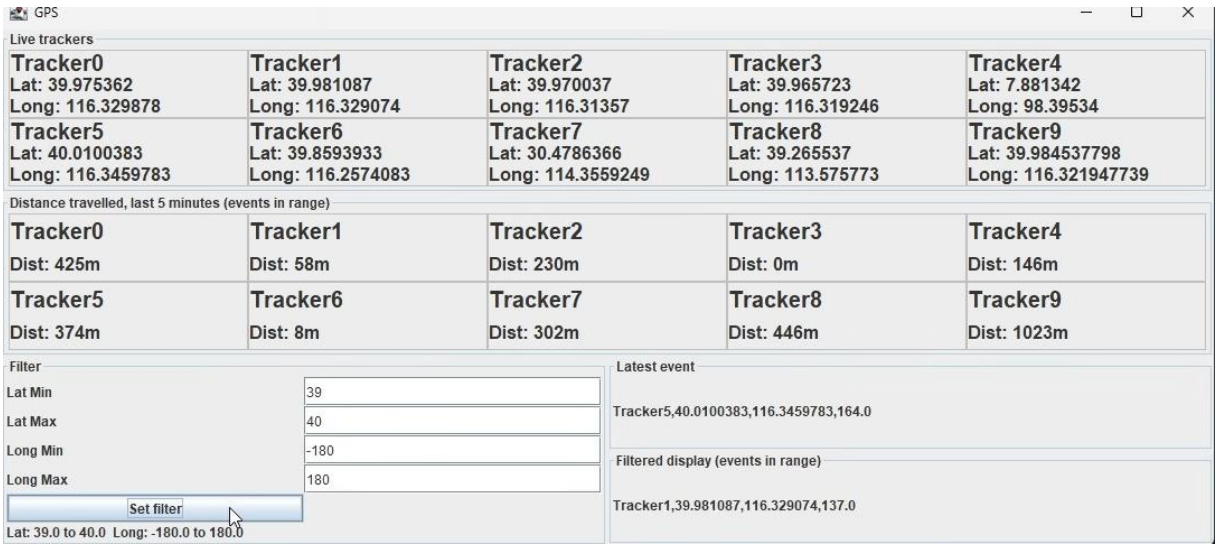
Live trackers: each of the ten trackers shows its tracker number, latitude and longitude, with no altitude.

Distance: the distance for each tracker over the last 5 minutes.

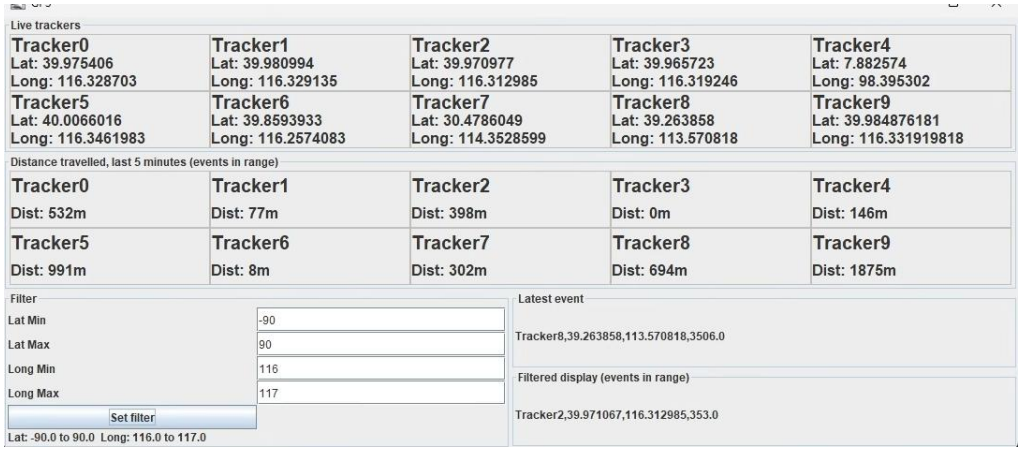
Filter boxes for latitude and longitude, a Set filter button and a label with the current range.

Latest event: shows all the four values of the most recent event.

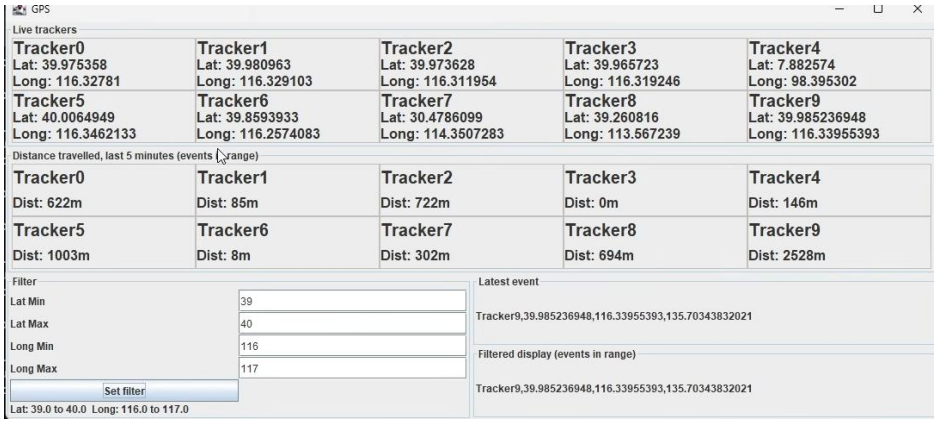
Filtered display: shows events inside the current range.



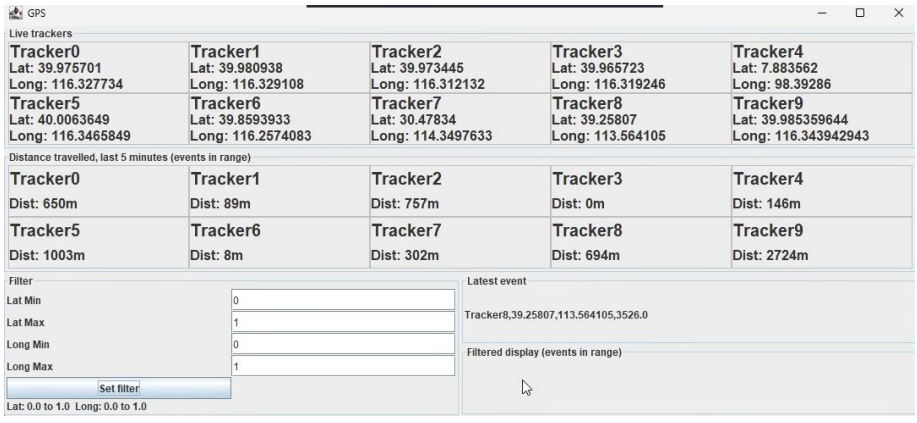
Set Latitude 39 to 40 to test the latitude function. Tracker0 (latitude about 39.98, inside the range) went from 374m to 425m. Tracker4 (latitude about 7.9, outside the range) stayed at 146m.



Set Longitude 116 to 117 to the longitude function. Tracker0 (longitude about 116.3, inside the range) went 477m to 532m. Tracker8 (longitude about 113.6, outside the range) stayed at 694m.



Set Latitude 39 to 40 and Longitude 116 to 117 together. Tracker0 and Tracker2 (inside both ranges) went from 568m to 622m and 542m to 722m. Tracker8 (outside the range) stayed at 694m.



Set Latitude 0 to 1 and Longitude 0 to 1 to test an empty result. No trackers fit in that range, so the Filtered display is empty while the field keeps updating.

Video link: https://youtu.be/fNkBTc57A8E?si=hZhPNOQ1U4hd1G02 (The video has the prior numbers that are not included here.)